

# L<sup>A</sup>T<sub>E</sub>X

## Introduction Course

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# Why LaTeX?

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- ▶ Open-Source and free
- ▶ High-quality typesetting
- ▶ Customizability
- ▶ Cross-platform
- ▶ Community and support
- ▶ Integration w other tools
- ▶ Overleaf

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Handle:

- ▶ Report
- ▶ Presentation
- ▶ Many more

# Code your document

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- ▶ Write your text
- ▶ Structure your document
- ▶ Create maths boxes
- ▶ Include images and tables
  - ▶ Size
  - ▶ Placement
  - ▶ Caption
  - ▶ Label and reference
- ▶ Create figure and graphes
- ▶ Bibliography
- ▶ Create your own commands and environments

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Know what you do  $\Rightarrow$  the document is as you want

# Overleaf

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Overleaf is a online editor:

- + No installation
- + Collaboration
- + Great UI
- + Lot of docs and templates
  - Limited free features
  - Internet connection
  - Dependency on a third party

# Huge corporation crashes, right?

May the 14th 2025, Overleaf servers were down for more than 6 hours, a lot of users were really frustrated, some panicked because of their deadlines, and in all this chaos, a lot of user were the happy ones, because they had the habit to have a local backup of their projects. To add to this, this was the day before the dealine for the NeurIPS 2025 and some Master thesis.

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Still need an IDE:

- ▶ TeXworks
- ▶ TeXstudio
- ▶ VSCode with LaTeX Workshop extension

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Complex setup that works:

- ▶ Vs code
- ▶ LaTeX Workshop extension
- ▶ Overleaf Workshop extension
- ▶ Overleaf cookies
- ▶ Local Git

# Writing

2 types of "text" boxes:

- ▶ Text box: for textual content (no delimiter)
- ▶ Math box: for mathematical content (delimiter:  $\backslash(\dots\backslash)$  or  $\$...\$$  for inline,  $\backslash[...\backslash]$  or  $\\$\\$\\$\\$$  for display)

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2 types of macros:

- ▶ Command: inline content, e.g. `\LaTeX`
- ▶ Environment: block of content with a begin and an end, e.g. `\begin{itemize} \dots \end{itemize}`

# Preamble

Before `\begin{document}`:

- ▶ Document class and options, `\documentclass{...}`
- ▶ Packages and their options, `\usepackage{...}`
- ▶ Custom commands and environments
- ▶ Document metadata, `\title{...}`, `\author{...}`,  
`\date{...}`

# Document management

Use the sectioning commands to structure your document:

- ▶ `\part{...}` for parts
- ▶ `\chapter{...}` for chapters (only in book and report document classes)
- ▶ `\section{...}` for sections
- ▶ `\subsection{...}` for subsections
- ▶ `\subsubsection{...}` for subsubsections
- ▶ `\paragraph{...}` for paragraphs
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Create a TOC with the `\tableofcontents`, `\listoffigures` and `\listoftables` commands (depth: `\setcounter{tocdepth}{...}`)

# Document management (cont.)

Reference:

- ▶ `\label{...}` command to label sections, figures, tables, equations, etc.
- ▶ `\ref{...}` command to reference the labeled items in the text
- ▶ `\pageref{...}` command to reference the page number of the labeled item

# Document management (cont.)

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## Multiple files management:

- ▶ Use the `\include{...}` command to include other .tex files in the main document
- ▶ Use the `\input{...}` command to include other .tex files without starting a new page

# Text box

Just write your text, and use command to format it:

- ▶ **Bold** with `\textbf{...}`
- ▶ *Italic* with `\textit{...}`
- ▶ Underline with `\underline{...}`
- ▶ Colored with `\color{red}...`

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You can also use some environments:

- ▶ Itemize for unordered list
- ▶ Enumerate for ordered list
- ▶ Description for description list
- ▶ And many more...

# Math box

Special characters and symbols:

- ▶ Greek letters: `\alpha`, `\beta`, ...
- ▶ Operators: `\sum`, `\int`, `\frac{\dots}{\dots}`, ...
- ▶ Relations: `\leq`, `\geq`, ...
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Some maths environments:

- ▶ Equation for numbered equations
- ▶ Align for aligned equations
- ▶ And many more...

For numbered maths environments, you can add a `*` to the end of the environment name to remove the numbering

# Figures and tables

Use the figure environment to include images:

- ▶ `\includegraphics{...}` to include the image
- ▶ `\caption{...}` to add a caption
- ▶ `\label{...}` to add a label for referencing

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Use the table environment to create tables:

- ▶ Use the tabular environment to define the structure of the table
- ▶ Use `&` to separate columns and `\\` to separate rows
- ▶ Use `\hline` to add horizontal lines
- ▶ Use `\caption{...}` and `\label{...}` for caption and referencing

array act as a math version of tabular

# Bibliography

Bibliography environment to create a bibliography:

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BibTeX or BibLaTeX for advanced management

# Create figure and graphes

TikZ package to create figures and graphes directly in LaTeX:

- ▶ `tikzpicture` environment to create a drawing area
- ▶ Use various TikZ commands to draw shapes, lines, and add text
- ▶ TikZ allows for complex and customizable figures, including flowcharts, diagrams, and more

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One of most complete packages, look the doc or search online

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For argument, use `[X]` for optional argument (in order) and `#X+n` for mandatory argument ( $X+n$  is the number of the argument)

## Issue with macros definition

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xparse package to create more user-friendly commands and environments:

- ▶ `\NewDocumentCommand{name}{arguments}{definition}`
- ▶ `\NewDocumentEnvironment{name}{arguments}{begin code}{end code}`

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Always have a local backup of your projects, especially if they are sensitive or have tight deadlines

# Practise time!

Create a new project on Overleaf and try to write a simple document using commands and macros (try some online search). You can also try to create a local LaTeX environment and link it to Overleaf using the method presented in this course. If any questions don't hesitate to ask me!